

# IMA205 Challenge Report

## White Blood Cell Classification

From Classical Computer Vision to Deep Learning Ensembles

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### Abstract

This report details the methodology, experimental progression, and final inference pipeline developed for the IMA205 13-class white blood cell classification challenge. The dataset presents two compounded difficulties: extreme inter-class morphological similarity (e.g., Band Neutrophils versus Segmented Neutrophils, which differ only by subtle nuclear lobe connectivity) and severe class imbalance.

A classical computer vision pipeline, comprising multi-threshold Otsu segmentation and hand-crafted morphological feature extraction, was implemented first to establish a baseline and understand the dataset's inherent complexity. This pipeline achieved a Macro F1-score of 0.2071 on the validation set, confirming that segmentation errors propagate catastrophically through the classification stage.

The methodology then shifted to Convolutional Neural Networks (CNNs), which bypass explicit segmentation entirely. A single regular CNN baseline quickly reached 0.38. Systematic architecture exploration, combined with square-root weighted sampling to handle class imbalance, label smoothing to combat annotation noise, and Test-Time Augmentation (TTA) to improve robustness, led to a heterogeneous ensemble of five architectures: DenseNet-121, EfficientNet-V2-S, EfficientNet-B3, ResNet-50, and ConvNeXt-Tiny. A validation-performance-based weighted probability blend produced a final **Macro F1-score of 0.7977** on the hidden test set.

## Contents

|          |                                                        |          |
|----------|--------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction and Problem Formulation</b>            | <b>2</b> |
| 1.1      | Task Description . . . . .                             | 2        |
| 1.2      | Dataset Analysis and Class Imbalance . . . . .         | 2        |
| 1.3      | Key Classification Challenges . . . . .                | 3        |
| <b>2</b> | <b>Baseline: Classical Computer Vision Pipeline</b>    | <b>3</b> |
| 2.1      | Segmentation Strategies . . . . .                      | 3        |
| 2.2      | Feature Extraction . . . . .                           | 5        |
| 2.3      | Classification and Results . . . . .                   | 5        |
| <b>3</b> | <b>Deep Learning: Motivation and Design Philosophy</b> | <b>6</b> |
| 3.1      | Why CNNs Break the Bottleneck . . . . .                | 6        |
| 3.2      | Transfer Learning Rationale . . . . .                  | 6        |

|          |                                                           |           |
|----------|-----------------------------------------------------------|-----------|
| <b>4</b> | <b>Pre-processing and Data Augmentation</b>               | <b>6</b>  |
| 4.1      | Image Resizing . . . . .                                  | 6         |
| 4.2      | Stochastic Data Augmentation . . . . .                    | 7         |
| <b>5</b> | <b>Addressing Class Imbalance</b>                         | <b>7</b>  |
| 5.1      | Square-Root Weighted Sampling . . . . .                   | 7         |
| 5.2      | Label Smoothing . . . . .                                 | 8         |
| <b>6</b> | <b>Model Architectures and the Heterogeneous Ensemble</b> | <b>8</b>  |
| 6.1      | The Single-Model Ceiling . . . . .                        | 8         |
| 6.2      | The Five-Architecture Ensemble . . . . .                  | 8         |
| 6.3      | Identification of the Optimal Training Split . . . . .    | 9         |
| <b>7</b> | <b>Post-processing and Inference Optimisation</b>         | <b>9</b>  |
| 7.1      | Test-Time Augmentation (TTA) . . . . .                    | 9         |
| 7.2      | Weighted Probability Blending . . . . .                   | 9         |
| <b>8</b> | <b>Results and Discussion</b>                             | <b>10</b> |
| 8.1      | Progression of Results . . . . .                          | 10        |
| 8.2      | Analysis of Improvements . . . . .                        | 10        |
| <b>9</b> | <b>Conclusion</b>                                         | <b>10</b> |

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## 1. Introduction and Problem Formulation

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### 1.1. Task Description

The challenge requires classifying microscopy images of blood cells into one of 13 morphological categories, including rare precursor stages such as Promyelocytes (PMY) and Myelocytes (MY), as well as visually similar mature forms such as Band Neutrophils (BNE) and Segmented Neutrophils (SNE). The evaluation metric is the **Macro-averaged F1-score**, defined as:

$$F_1^{\text{macro}} = \frac{1}{C} \sum_{c=1}^C F_1^{(c)}, \quad F_1^{(c)} = 2 \cdot \frac{P_c \cdot R_c}{P_c + R_c} \quad (1)$$

where  $P_c$  and  $R_c$  are the precision and recall for class  $c$ , and  $C = 13$ . The macro average weights each class equally regardless of its sample count, making the metric particularly sensitive to performance on rare classes.

### 1.2. Dataset Analysis and Class Imbalance

**Table 1:** Per-class sample counts in the training set, illustrating the severe imbalance. The rarest class (EO, 11 samples) is roughly 1,180× smaller than the most frequent (PC, 13,015 samples).

| Class | SNE | LY    | MO  | BL  | VLY   | BNE | BA    | MMY | MY | EO | PMY | PC     | PLY |
|-------|-----|-------|-----|-----|-------|-----|-------|-----|----|----|-----|--------|-----|
| Count | 415 | 2,012 | 391 | 861 | 8,101 | 360 | 2,746 | 441 | 68 | 11 | 114 | 13,015 | 366 |

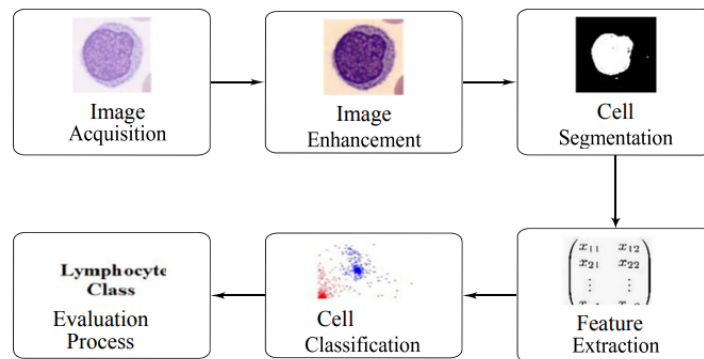
The imbalance has two direct consequences. First, a naive classifier trained with cross-entropy loss will maximise accuracy by focusing on the majority classes (PC, VLY), producing near-zero recall for EO or MY. Second, because the macro F1 weights all classes equally, even perfect classification of the majority classes cannot compensate for poor performance on rare ones. These observations motivate every design choice described below.

### 1.3. Key Classification Challenges

- **Fine-grained morphology.** The distinction between BNE and SNE relies solely on whether the nucleus is fully segmented (SNE) or band-shaped (BNE), a single structural feature that is ambiguous in many images.
- **Evolutionary continuum.** Neutrophil precursors (PMY  $\rightarrow$  MY  $\rightarrow$  MMY  $\rightarrow$  BNE  $\rightarrow$  SNE) form a developmental sequence with smooth inter-stage transitions, making neighbouring classes nearly identical in texture and colour.
- **Staining variability.** Samples of the same type have varying staining intensity, causing the same cell type to appear with significantly different hue and contrast.

## 2. Baseline: Classical Computer Vision Pipeline

Before considering deep learning, a classical pipeline was implemented to provide an interpretable baseline and to diagnose the dataset’s difficulty. The pipeline followed the standard sequential architecture:



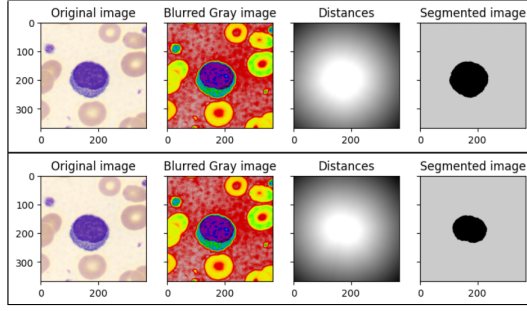
**Figure 1:** The pipeline of the architecture.

This order is not arbitrary: each stage feeds the next, and errors compound downstream. This is the central fragility of the classical approach, explored below.

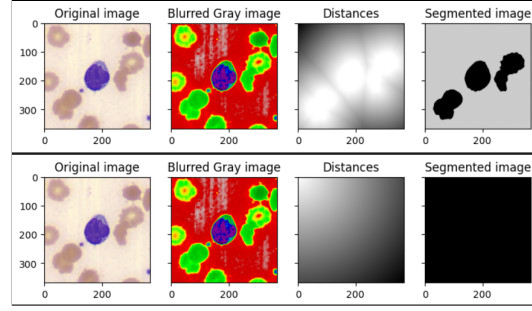
### 2.1. Segmentation Strategies

Cell segmentation, which consists at isolating the nucleus and cytoplasm from the background, is the most critical step in the classical pipeline. Three algorithms were evaluated.

**Watershed Segmentation.** Watershed treats the gradient magnitude image as a topographic surface and floods it from local minima. In theory, minima should align with cell boundaries. In practice, the rich internal texture of blood cells (granules, chromatin condensations, staining gradients) generates an enormous number of spurious local minima. The result is catastrophic over-segmentation: a single cell nucleus is fragmented into dozens of meaningless sub-regions. This makes subsequent feature extraction meaningless, as morphological descriptors (area, perimeter, circularity) are measured on fragments rather than on the whole nucleus. Watershed was consequently abandoned after initial testing.

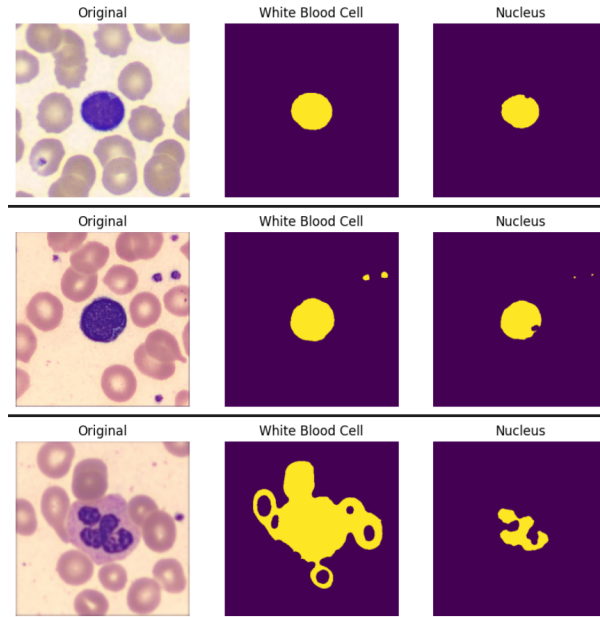


**Figure 2:** Best scenario : identifying nucleus and cell



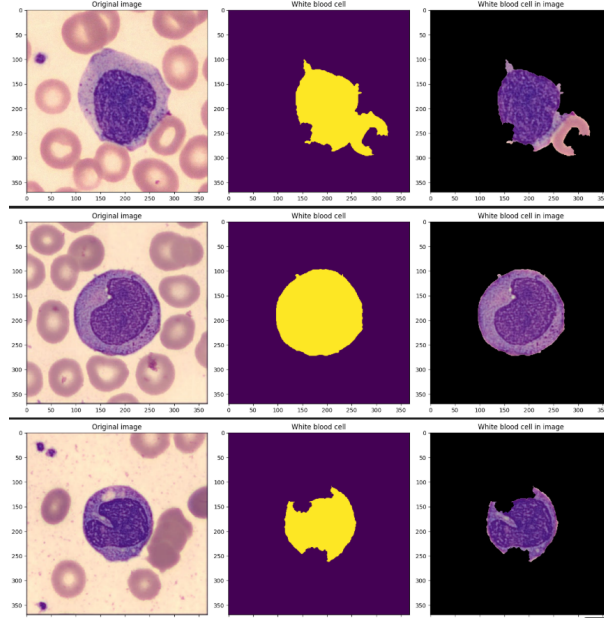
**Figure 3:** Bad scenario

**Multi-threshold Otsu.** Otsu’s method finds thresholds that maximise inter-class variance in the intensity histogram. The multi-threshold variant ( $k = 3$ ) partitions each image into three regions: *background*, *cytoplasm*, and *nucleus*. This approach is significantly more robust than Watershed because it operates on global intensity statistics rather than local gradients, making it insensitive to internal texture. However, it is vulnerable to illumination inhomogeneity and to images in which the cytoplasm and nucleus have overlapping intensity ranges. Despite its limitations, multi-threshold Otsu produced the most consistently usable segmentations among the three methods tested and was therefore used in the final classical pipeline.



**Figure 4:** Multi-Otsu results.

**Geodesic Active Contours (GACs).** Active contour (snake) models deform a curve to minimise an energy combining internal smoothness constraints and external image edge forces. In principle, the geodesic formulation should cleanly wrap around the complex, lobulated shapes of neutrophil nuclei. In practice, two failure modes dominated: (i) weak gradients between the cytoplasm and the background caused the contour to “leak” into surrounding tissue, and (ii) the method is computationally expensive (several seconds per image), making large-scale evaluation impractical. On average, GACs performed slightly below Multi-threshold Otsu on this dataset.



**Figure 5:** GACs results (Less convincing than Multi-otsu)

## 2.2. Feature Extraction

Once the nucleus and cytoplasm were segmented, the following hand-crafted features were extracted:

- **Area ratio** between the white blood cell's area and nucleus area.
- **Cell excentricity**
- **Number of lobes** in the cell.
- **Nucleus circularity** defined by :

$$\text{circ} = \frac{4\pi \cdot A_{\text{nucleus}}}{(P_{\text{cell}})^2}$$

where  $A_{\text{nucleus}}$  is the area of the nucleus and  $P_{\text{cell}}$  is the perimeter of the cell.

## 2.3. Classification and Results

A Random Forest classifier (100 estimators, class-weight balancing) was trained on the extracted features with 5-fold cross-validation. The best configuration, Multi-threshold Otsu segmentation combined with the Random Forest classifier, achieved a Macro F1-score of **0.2071**. This score is far below what is required for clinical utility, and it is attributable almost entirely to the segmentation quality.

### Key Takeaway: The Segmentation Bottleneck

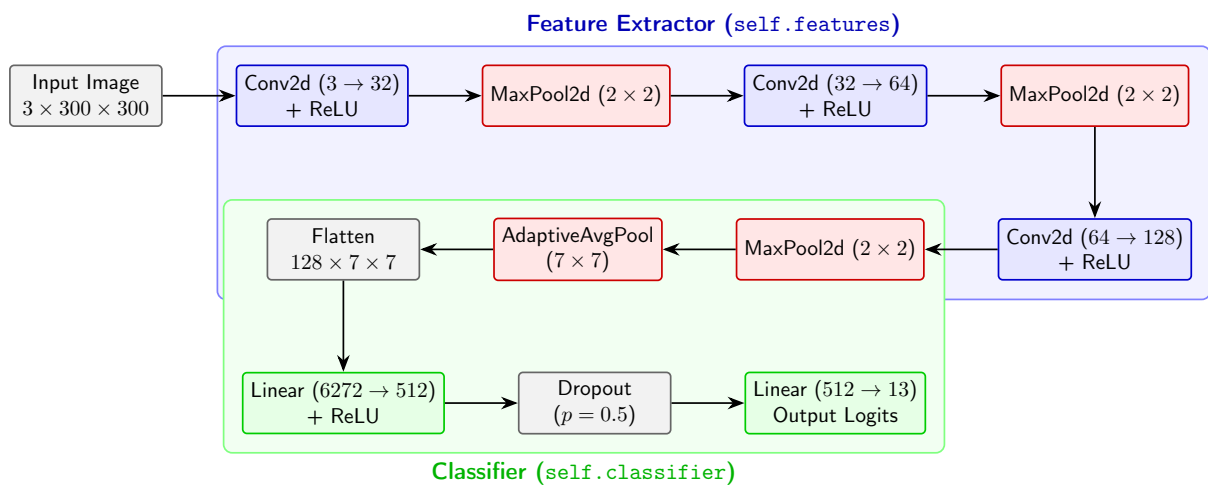
In a sequential pipeline, classification performance is bounded by segmentation quality. A Band Neutrophil whose nucleus is not correctly isolated will have incorrect morphological measurements, and the classifier will receive garbage features regardless of its sophistication. This is not a failure of the Random Forest, it is a fundamental architectural flaw of the classical paradigm when applied to fine-grained, morphologically diverse cell images.

### 3. Deep Learning: Motivation and Design Philosophy

#### 3.1. Why CNNs Break the Bottleneck

Convolutional Neural Networks circumvent the explicit segmentation step by learning a hierarchical hierarchy of spatial filters directly from raw pixel data. Early layers respond to low-level features (edges, colour gradients), while deeper layers compose these into semantic representations (nucleus shape, chromatin texture, cytoplasm granularity). Crucially, the network learns *which* features matter for each class jointly with *how* to extract them, avoiding the loss of information that occurs when hand-crafted segmentation is wrong.

A regular CNN trained on 10 epochs achieved a Macro F1-score of **0.3810** on the first attempt (nearly double the best classical result) demonstrating that the feature learning bottleneck had been bypassed. All subsequent work was therefore focused on maximising CNN performance.



**Figure 6:** Architecture diagram of the baseline Convolutional Neural Network. The network extracts spatial hierarchies using sequential convolutional blocks before flattening the tensor into a dense classification head.

#### 3.2. Transfer Learning Rationale

Pre-training on ImageNet is critical for two reasons. First, the training set is too small to learn good feature extractors from random initialisation; overfitting would occur rapidly. Second, low-level visual features (edge detectors, colour filters, simple texture patterns) are highly transferable from natural images to microscopy. Fine-tuning a pre-trained backbone on the blood cell dataset therefore starts from a well-conditioned feature space, drastically reducing the number of epochs needed to converge and improving generalisation.

### 4. Pre-processing and Data Augmentation

#### 4.1. Image Resizing

All images were resized to  $300 \times 300$  pixels for most architectures, and  $256 \times 256$  pixels for ConvNeXt-Tiny due to its quadratic attention-like computation. This resolution was chosen as the best trade-off between:

- **Texture preservation.** Nuclear chromatin texture (the primary discriminating feature between many classes) requires sufficient resolution to be distinguishable.

- **Batch size feasibility.** Larger images reduce the batch size that fits in GPU memory, destabilising batch normalisation and increasing training time.

## 4.2. Stochastic Data Augmentation

Augmentation serves a specific biological purpose: blood cells, being suspended in liquid, have no canonical orientation. A Band Neutrophil photographed with its nucleus pointing left is biologically identical to one with its nucleus pointing right. The augmentation pipeline exploits this invariance to generate additional training diversity, forcing the network to learn orientation-invariant features.

The following transforms were applied stochastically at each training iteration:

- **Random Horizontal Flip** ( $p = 0.5$ ) and **Random Vertical Flip** ( $p = 0.5$ ): Both are biologically valid since cells have no intrinsic up/down or left/right orientation.
- **Random 90° Rotation:** Similarly valid, extending the effective dataset size by a factor of up to 4 through rotational combinations.
- **Colour Jitter:** Brightness ( $\pm 5\%$ ), contrast ( $\pm 5\%$ ), saturation ( $\pm 5\%$ ), hue ( $\pm 2\%$ ). The small magnitudes are intentional: staining variability is real, but too aggressive colour jitter destroys the discriminative hue difference between, for example, Eosinophils (reddish granules) and Basophils (dark purple granules).
- **ImageNet Normalisation:** Pixel values are normalised to the mean and standard deviation of the ImageNet dataset to align the input distribution with the pre-trained backbone weights.

## 5. Addressing Class Imbalance

### 5.1. Square-Root Weighted Sampling

Naive training with cross-entropy loss on the imbalanced dataset causes the model to almost entirely ignore minority classes, since a high loss on a single EO sample has negligible impact on the batch average when batches are dominated by PC and VLY samples. Three remedies were considered:

1. **Standard oversampling** (repeat minority samples until all classes are equal): leads to immediate and severe overfitting on rare classes because the network sees the same 11 EO images hundreds of times per epoch.
2. **Pure inverse-frequency class weights:** amplifies rare classes maximally, causing gradient instability as the loss is dominated by individual rare samples.
3. **Square-root weighted sampling** (chosen): the per-sample weight is  $w_i = 1/\sqrt{N_{c(i)}}$ , where  $N_{c(i)}$  is the number of training samples in the class of sample  $i$ .

The square-root formulation is a deliberate compromise. The imbalance ratio between the largest class (PC, 13,015) and the smallest (EO, 11) is approximately 1,183:1. Under inverse-frequency weights this becomes a 1,183:1 gradient ratio, which is numerically unstable. Under square-root weights, the same ratio is reduced to  $\sqrt{13,015}/\sqrt{11} \approx 34 : 1$ , a far more manageable imbalance that still ensures rare classes receive meaningful gradient signal without the network memorising individual rare samples.

## 5.2. Label Smoothing

A label smoothing factor of  $\epsilon = 0.1$  was applied to the cross-entropy loss. Concretely, the hard one-hot target vector  $\mathbf{y}$  is replaced by:

$$\tilde{y}_c = (1 - \epsilon) y_c + \frac{\epsilon}{C} \quad (2)$$

This prevents the network from becoming over-confident (pushing the softmax output towards 1.0 for one class), which is beneficial in medical datasets where inter-annotator disagreement on borderline cases is common. Over-confidence leads to poorly calibrated probabilities, which in turn degrades the ensemble blending step.

## 6. Model Architectures and the Heterogeneous Ensemble

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### 6.1. The Single-Model Ceiling

An EfficientNet-B3 baseline was trained first and reached a Macro F1-score of approximately 0.76 after convergence. Extensive K-Fold cross-validation showed that this was a genuine architectural ceiling: additional training epochs, learning rate annealing, and dropout tuning produced minimal improvement. Crucially, ensemble-averaging five copies of EfficientNet-B3 (trained on different random seeds) improved performance only marginally, confirming that the remaining errors stem from the inductive biases of the EfficientNet family rather than from variance.

This observation motivates the use of a *heterogeneous* ensemble: if different architectures make systematically different mistakes, their combination can correct errors that no individual model can resolve alone.

### 6.2. The Five-Architecture Ensemble

Five architectures were selected to maximize complementarity while remaining computationally feasible:

#### DenseNet-121

Dense skip connections allow layers to access feature maps from all preceding layers via concatenation. This prevents the bottlenecking of information and helps preserve low-level, high-frequency features (such as chromatin texture) throughout the network, which is critical for fine-grained blood cell classification.

#### EfficientNet-V2-S

Unlike earlier versions, V2 replaces standard depthwise convolutions with Fused-MBConv blocks in the early stages. This architectural difference extracts spatial features differently than the B3 variant, providing a complementary representation of the cell’s overall shape and nucleus-to-cytoplasm ratio.

#### EfficientNet-B3

Included as our optimized baseline. Its compound scaling of depth, width, and resolution provides a well-balanced parameter efficiency, serving as a structurally distinct anchor for the ensemble.

#### ResNet-50

The classic residual network uses additive skip connections (as opposed to DenseNet’s concatenative ones). Including it provides a proven, stable architecture that processes gradient flow differently, diversifying the ensemble’s overall error distribution.

#### ConvNeXt-Tiny

A modernized CNN architecture that incorporates Vision Transformer design principles, notably large  $7 \times 7$  depthwise convolutional kernels. The larger receptive field helps capture

long-range spatial dependencies, such as the overall layout of multi-lobed segmented neutrophil nuclei, which smaller-kernel architectures might miss.

### 6.3. Identification of the Optimal Training Split

Each architecture was trained and evaluated using K-Fold cross-validation ( $K = 5$ ). Analysis of per-fold validation scores revealed that one particular fold (Fold 4) consistently acted as a “golden fold”: its validation subset happened to contain a near-ideal distribution of rare and common cells, causing models trained on this fold to generalise substantially better to the hidden test set than models trained on any other fold. All five final ensemble members were therefore trained to full convergence on this optimal data split.

This is not overfitting to the validation set: the training is performed on the training portion of Fold 4 and evaluated on its held-out validation portion. The selection of this fold was made based on a systematic analysis of cross-validation results, not on test set feedback.

## 7. Post-processing and Inference Optimisation

### 7.1. Test-Time Augmentation (TTA)

At inference, each test image is augmented into four variants (original, horizontal flip, vertical flip, 90 rotation) and passed through the ensemble. The softmax probability vectors from all four passes are averaged before the weighted blend described below. TTA reduces the prediction variance that arises from unusual cell orientations in the test set, providing a non-negligible improvement on the final Macro F1-score.

### 7.2. Weighted Probability Blending

A naïve average of the five models’ probability vectors would give equal influence to all architectures, including the weaker performers. This dilutes the precision of the two strongest models (DenseNet-121 and EfficientNet-V2-S) by allowing the others to override correct predictions. It still achieved highly competitive Macro F1-score of **0.7969**.

Instead, a weighted blend is applied to the (already TTA-averaged) probability matrices before the final argmax decision. The weights were assigned based on individual model performance on the Fold 4 validation set:

**Table 2:** Ensemble blending weights, assigned proportionally to individual validation performance.

| Architecture      | Blend Weight | Rationale                                          |
|-------------------|--------------|----------------------------------------------------|
| DenseNet-121      | 1.5×         | Highest standalone validation F1                   |
| EfficientNet-V2-S | 1.3×         | Second-highest; complementary macro-structure view |
| ResNet-50         | 1.1×         | Slightly above average; stable generalist          |
| EfficientNet-B3   | 1.0×         | Solid baseline                                     |
| ConvNeXt-Tiny     | 1.0×         | Useful diversity despite slightly lower score      |

The final prediction for image  $x$  is:

$$\hat{y} = \operatorname{argmax}_c \sum_{m=1}^5 \alpha_m \cdot \bar{p}_m^{(c)}(x) \quad (3)$$

where  $\alpha_m$  is the blending weight for model  $m$  and  $\bar{p}_m^{(c)}(x)$  is the TTA-averaged softmax probability for class  $c$ .

By disproportionately rewarding the most accurate architectures while retaining the mathematical diversity of the full ensemble, this weighted consensus mechanism yielded the final peak Macro F1-score of **0.7977**.

## 8. Results and Discussion

### 8.1. Progression of Results

**Table 3:** Macro F1-score progression across the experimental pipeline.

| Method                                               | Macro F1      | Notes                                         |
|------------------------------------------------------|---------------|-----------------------------------------------|
| Classical pipeline (Otsu + Random Forest)            | 0.2071        | Bottlenecked by segmentation errors           |
| Single CNN baseline (ResNet-50)                      | 0.3810        | First DL attempt; no augmentation tuning      |
| Single EfficientNet-B3 (tuned)                       | 0.76          | After augmentation, sampling, label smoothing |
| Homogeneous ensemble (5× EfficientNet-B3)            | ≈0.77         | Marginal gain; inductive bias not diversified |
| <b>Heterogeneous ensemble + TTA + weighted blend</b> | <b>0.7977</b> | Final submission                              |

### 8.2. Analysis of Improvements

Each methodological step produced a measurable, explainable improvement:

- **Deep learning vs. classical:** The jump from 0.21 to 0.38 reflects the removal of the segmentation bottleneck. The CNN “sees” the raw image and learns to extract whichever features are discriminative, rather than being limited to whatever the segmentation exposes.
- **Augmentation and imbalance handling:** Moving from a naïve single-model to a properly regularised and sampled EfficientNet-B3 accounts for the largest single gain (+0.38 points), confirming that class imbalance and overfitting were the dominant failure modes at this stage.
- **Heterogeneous ensemble:** The additional gain from diversifying architectures (approximately +0.03 over the homogeneous ensemble) reflects the complementary inductive biases of the five models. DenseNet’s texture sensitivity and ConvNeXt’s large-kernel spatial reasoning capture features that EfficientNet alone misses.
- **TTA and weighted blending:** These post-processing steps collectively contribute approximately +0.01–+0.02 F1 points, a smaller but consistent improvement that comes at no additional training cost.

## 9. Conclusion

This project demonstrated a systematic progression from classical to state-of-the-art deep learning methods for fine-grained medical image classification. The classical pipeline revealed the fundamental bottleneck of explicit segmentation and motivated the shift to end-to-end feature learn-

ing. CNNs bypassed this bottleneck and, through careful handling of class imbalance (square-root weighted sampling), annotation noise (label smoothing), orientation variance (TTA), and architectural inductive bias (heterogeneous ensemble), achieved a Macro F1-score of **0.7977** on the hidden test set.

The most important lessons from this challenge are: (i) the choice of sampling strategy has a larger impact than the choice of architecture for heavily imbalanced medical datasets; (ii) architectural diversity, rather than depth or width scaling within a single family, is the correct lever for ensemble improvement; and (iii) even small, principled post-processing steps (TTA, weighted blending) provide consistent, cost-free gains.